

Miguel Aenlle

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EDUCATION

University of Illinois at Urbana-Champaign

Bachelor of Science in Computer Science

Expected: May 2027

Cumulative GPA: 4.0/4.0

- Honors: [Provost Scholarship \(Full-tuition Scholarship\)](#); [UIUC Research Park Most Outstanding Intern Award \(1 of 800 selected\)](#); [Engineering Visionary Scholarship](#); [GFX Innovation Scholar](#); [James Scholar](#); [Dean's List 2024/2025](#)
- Relevant Coursework: Distributed Systems, Cloud Computing, Systems Programming, Database Systems, Data Structures and Algorithms, Deep Learning, Computer Vision, Reinforcement Learning, C++, Java

PROFESSIONAL EXPERIENCE

Software Engineering Intern, PrairieLearn, Inc.

Aug. 2025 – Present

- Accelerated manual grading process from **1 week to 1 hour** by creating LLM-driven handwriting grading pipeline
- Architected backend scaling to grade **1,800 submissions/hr** per course with Python, TypeScript, Redis, & SQL
- Achieved human-level **98%** grading accuracy using GCP, Gemini, prompt-tuning experiments, & image correction
- Created staff dashboard visualizing real-time progress of **100s** of concurrent grading jobs with React & SocketIO

Software Engineering Intern, PrairieLearn, Inc.

Dec. 2024 – Aug. 2025

- Engineered [image-capture API](#) delivering **1ms** captures for **180,000** students in high-stakes exams & assessments
- Redesigned [staff navigation system](#) across **80+** pages, enhancing experience for over **2,000** course instructors
- Implemented UIs and API endpoints with TypeScript, React, HTML, CSS, WebSockets, PostgreSQL, and Python

Software Engineering Intern, PATHWAYOs

Jun. 2023 – Aug. 2023

- Created work-based learning experience matching system for **500,000**-student platform with React & TypeScript
- Built low-latency Swift/MySQL API for opportunity matching via preferences, location, experience, & availability
- Iterated features based on feedback and peer reviews from engineering and product teams with GitHub and Slack

Software Developer, Normandy Remodeling

Aug. 2022 – Jan. 2025

- Architected [cloud-based platform](#) managing logistical and financial data on **33,000** customers and **9,900** projects
- Built frontend with React, CSS, and TypeScript, and backend with Node.js, Express, MongoDB, Jest, and Azure
- Automated document, email, and invoicing workflows for **100 employees** via Azure & OneDrive/Office 365 APIs

RESEARCH EXPERIENCE

Software Developer, Google Summer of Code – Emory University

Jun. 2025 – Present

- Engineered [ML pipeline](#) to train and optimize automated MRI segmentation models with OpenCV and PyTorch
- Expanded limited proprietary training data by **1,600%** via custom-trained, state-of-the-art generative AI models
- Improved segmentation accuracy from **39%** to **80%**. Collaborated with Google and Emory University researchers

Undergraduate Researcher, University of Illinois Urbana-Champaign

Oct. 2024 – May 2025

- Developed [AI model](#) to inject imperceptible noise into audio, disrupting malicious audio deepfake attacks
- Implemented GPU-accelerated PyTorch pipeline for training and optimizing the model on **1,000+** speakers
- Collaborated with UIUC, Google, and Meta AI researchers to develop the audio defense technologies

LEADERSHIP AND ACTIVITIES

Web Team Lead, HackIllinois

Jul. 2025 – Present

- Led redesign of [site](#), [registration](#), and [admin web systems](#) for UIUC's **1,000+** attendee flagship hackathon
- Revamped admin site, implementing features to streamline applicant, event, and sponsor data management
- Engineered web-based platform to automate corporate outreach task management across over **500** contacts

Software Developer, Reflections Projections

Mar. 2025 – Sept. 2025

- Revamped [sponsor resumebook](#), enhancing search, secure export, and rapid navigation of **2,000+** resumes
- Developed interactive, bespoke schedule, speaker, and Q&A pages for Midwest's largest collegiate tech conference

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C++, SQL, Swift, Java, HTML, CSS, Dart, MIPS, Verilog

Web and Backend: React, NextJS, NodeJS, Express, REST, WebSockets, MongoDB, MySQL, PostgreSQL, Redis

Data Science and AI: PyTorch, OpenCV, Google Gemini, OpenAI, NumPy, Pandas, Matplotlib, spaCy NLP, NLTK

Other: Git, GitHub, CI/CD, GitHub Actions, Firebase, Google Cloud, Jest, Agile, Linux, SwiftUI, Flutter, Azure